Eben Radocchia

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EDUCATION

University of Vermont (UVM), Burlington, VT — Expected Graduation: June 2029
Bachelor of Science in Electrical Engineering — *GPA: 3.98 (Major: 4.0)*
Undergraduate Certificate in Autonomy and Robotics

- **Related Courses:** Circuits I, Signals & Systems, Computer Organization.
- **Honors & Awards:** UVM Green & Gold Scholarship (2025), Rockowitz Scholarship (2025).

TECHNICAL SKILLS

Languages: Python, C/C++, Java, LaTeX
Tools: ROS 2, LTspice, ngspice, KiCad, AutoCAD, Linux
Hardware: CAN bus, Wiring harnesses, Electronic troubleshooting, Test equipment

RELATED EXPERIENCE

Alternative Energy Racing Organization (AERO), Burlington, VT — November 2025 – Present
Inverter Project Lead / Electrical Team

- Lead motor and inverter integration for the 2026–27 season, including a new DTI HV-550 inverter.
- On the battery pack team, measured cell voltages, insulated modules with Kapton for safe HV storage, and helped build a minipack for testing.
- Flashed firmware on a Cascadia Motion PM100 inverter, built its wiring harnesses, and calibrated the Emrax 228 motor's resolver offset using back-EMF coast-down measurements.
- Configured CAN communication with the inverter and monitored live telemetry during motor testing.
- Achieved the team's first powered wheel spin in three years at the 2026 competition.

UVM Research Experience for Undergraduates, Burlington, VT — September 2026 – Present
Undergraduate Researcher

- Developing UVM's Autonomous Weeding Robot under Dr. Luis Duffaut, redesigning the drivetrain for rougher terrain and an adjustable track width to match crop row spacing.
- Built ROS 2 simulations of a differential drive robot, including a PD go-to-goal controller and a potential field planner using simulated LIDAR, visualized in Gazebo and RViz.

FIRST Tech Challenge Team 18650, Bennington, VT — September 2025 – Present
Mentor

- Mentor students on debugging Java robot code, sensing, and autonomous routines.
- Attended the 2026 FIRST World Championship with Team 18650, the first southern VT team to qualify.

Team Captain — September 2024 – June 2025

- Led team in the 2024–2025 season to the winning alliance at the Vermont State Championship, placing 2nd overall, and securing an invitation to the New England Premier Event.

PROJECTS

Polyphonic Digital Synthesizer Keyboard — Spring 2026 – Present

- Built a 13-key, velocity-sensitive, 8-voice synthesizer on an ATmega328P (team of 2) using direct digital synthesis, with force sensors multiplexed into one ADC and register-level timer/PWM audio output.
- Designing a custom KiCad PCB to replace the breadboard prototype.

ADDITIONAL EXPERIENCE

Nature's Market, Manchester, VT — 2022 – August 2025, Summer 2026
Cashier and Trainer

- Trained 3 new employees on store policies, inventory, and customer service.

Sonatina Piano Camp, Bennington, VT — Summers 2025–2026
Junior Faculty Member

- Taught private piano lessons to 7 students while supervising and supporting 20+ campers of all ages.